SGA 5: ℓ-adic Cohomology and L-Functions

Draft English translation of the volume introduction

Introduction

This volume brings together the exposés from Grothendieck's seminar at the IHES in 1965–66, initially distributed in the form of mimeographed notes. Compared with the original version, the only important changes concern Exposé II, which is not reproduced, and Exposé III, which has been entirely rewritten and enlarged by an appendix numbered III B. Apart from a few detailed modifications and added footnotes, the other exposés have been left as they stood.

The heart of the seminar is the Lefschetz formula in étale cohomology (III, III B, XII), and its application to the cohomological interpretation of L-functions (XIV). All the results announced by Grothendieck in his Bourbaki seminar talk [2] are proved here completely. The trace formulae established in (III, III B, and XII), by different methods, are more general than is necessary for proving merely the rationality of the L-functions. A complete and much shorter proof of the latter appears in Deligne's exposé (SGA $4^{1/2}$, Report), where Grothendieck's method from Exposés XII and XIV is followed, but stripped of all superfluous generality. We nevertheless hope that the formulae of III and III B may be useful in other situations. As for the rest of the seminar, it consists of two exposés on the theory of passage to the limit that gives rise to ℓ-adic cohomology (V, VI), and of various complements to the duality formalism (I, VII) and to the Lefschetz formula (VII, X).

Here, more precisely, is the content of this volume.

Exposé I is independent of the rest of the seminar. Its main result is that, on a regular scheme satisfying certain additional local hypotheses, and assuming the availability of resolution of singularities and the purity theorem, the constant sheaf with value $\mathbb{Z}/n\mathbb{Z}$, for n prime to the residue characteristics, is dualizing (I.3.4.1). This theorem applies in particular to excellent regular schemes of characteristic zero, thanks to Hironaka and to Artin's results (SGA 4, XIX). The case of a regular scheme of dimension 1 is treated separately, by another, very elementary method; see also SGA $4^{1/2}$, Duality. Let us also point out that, by a different argument using neither resolution nor purity, Deligne proves in SGA $4^{1/2}$, Finiteness Theorem, that if $f : X \to S$ is a scheme of finite type over a regular scheme of dimension 0 or 1, then the complex $f^!(\mathbb{Z}/n\mathbb{Z})_S$, for n prime to the residue characteristics, is dualizing. It is not known, however, whether this result, which is more useful than the biduality theorem of Exposé I, and which overlaps it without containing it, extends to the case where the base scheme S is excellent, regular, and of dimension > 1.

As was said above, Exposé II, entitled "Künneth formulae for cohomology with arbitrary supports", does not appear in this volume. Written by L. Illusie from Grothendieck's handwritten notes, it was devoted to theorems on cohomological properness and generic local acyclicity; but these were proved only under resolution hypotheses. A proof of the same results without these hypotheses, obtained since then by Deligne (SGA $4^{1/2}$, Finiteness Theorem), made publication of that exposé unnecessary. Certain complements, concerning for example the case of normal-crossing divisors, were incorporated into an appendix to *loc. cit.* We add that theorems analogous to those of *loc. cit.*, in the case of sheaves of sets and noncommutative groups, are proved in Mme Raynaud's exposé (SGA 1, XIII).

Exposé III contains a proof of the Lefschetz–Verdier formula for cohomological correspondences between complexes of sheaves on schemes of finite type over a field. In Exposé III B, one shows how to compute the local terms of the formula in certain cases, for instance that of the Frobenius correspondence on curves. We refer the reader to the introductions of Exposés III and III B for more details on their contents.

Exposé IV, by A. Grothendieck, devoted to the construction of the cohomology class associated with a cycle, was written up by P. Deligne and published in SGA $4^{1/2}$.

After technical preliminaries in Exposé V on adic inverse systems, Exposé VI defines the notion of a constructible $\mathbb{Z}_\ell$-sheaf, respectively a constructible twisted constant sheaf, that is, a "smooth" sheaf in Deligne's terminology ([1], SGA $4^{1/2}$, Arcata). It then extends to constructible $\mathbb{Z}_\ell$-sheaves, by passage to the limit from $\mathbb{Z}/\ell^n$-sheaves, the formalism of higher direct images with proper supports. Exposé VI does not, however, give a definition of a derived category of $\mathbb{Z}_\ell$-sheaves, with operations $\otimes^{\mathrm{L}}$ and RHom extending those available in the derived category of $\mathbb{Z}/\ell^n$-sheaves. A fortiori, the extension to $\mathbb{Z}_\ell$-sheaves of the duality formalism of SGA 4, XVIII is not considered. This question was the subject of Jouanolou's unpublished thesis. The reader may nevertheless consult the beginning of [1], where the sheaves $\underline{\mathrm{Ext}}^i$ of $\mathbb{Z}_\ell$-sheaves are defined.

Exposé VII, independent of the rest of the seminar, is undoubtedly one of the most interesting. It contains: (a) the computation of the cohomology of several standard varieties, namely affine spaces, projective spaces, and flag varieties; (b) an account of the theory of Chern classes in étale cohomology; (c) a proof of the self-intersection formula

$$i^* i_* x = x\, c_d(N)$$

in étale cohomology, respectively in the Chow ring, the proof in the latter case being due to Mumford, together with various applications of this formula, including computation of the cohomology of blow-ups and the Gauss–Bonnet formula.

Exposé VIII contains sorites on finiteness conditions in derived categories and Grothendieck groups. These questions are taken up again in a more general framework in SGA 6, I and IV.

Exposé IX, by J.-P. Serre, entitled "Introduction to Brauer theory", was published in [4] and is not reproduced in this volume. One of its principal results, due to Swan, concerning the existence of a projective module having as character the "Swan character", is used in Exposé X, where an Euler–Poincaré formula is proved for a sheaf on a curve, slightly more general than the one appearing in Raynaud's exposé [3].

Exposé XI, written by I. Bucur, does not exist: it was lost during a move, and the author had no copy of it. It was devoted to Grothendieck's theory of commutative traces, generalizing that of Stallings [5]. A more sophisticated version of this theory, in a sheaf-theoretic framework, is presented in (III B 5, 6). Applications to Lefschetz formulae are given in Exposés XII and III B. The trace formula of Exposé XII is proved independently of the general formula of Exposé III, but it is shown in (III B 6) that the local terms appearing there are indeed those of the general formula, and that the latter implies it.

The absence of Exposé XIII is due only to numbering: the contents of Exposés X through XII had initially been planned to extend over four exposés.

Exposé XIV, sometimes also numbered XV, contains generalities on the Frobenius correspondence, including in particular the comparison between "geometric" Frobenius and "arithmetic" Frobenius. Starting from the trace formula of Exposé XII, or from III B, it gives Grothendieck's proof of the rationality of L-functions. The two important points are reduction to the case of curves, and the passage to the limit that makes it possible to reduce to a formula for $\mathbb{Z}/\ell^n$-sheaves.

During the oral seminar, Grothendieck gave an exposé on open problems and stated a few conjectures. That exposé, which was intended to close the seminar, was unfortunately not written up, nor was his very beautiful introductory exposé, which surveyed the Euler–Poincaré and Lefschetz formulae in various contexts: topological, complex analytic, and algebraic.

I thank P. Deligne for having convinced me to write, in a new version of Exposé III, a proof of the Lefschetz–Verdier formula, thereby removing one of the obstacles to the publication of this seminar. I am very grateful to him for the improvements in exposition that he suggested to me, and for the help he gave me in preparing this volume for the publisher. I also thank Mme Lièvremont, who typed Exposés III and III B carefully and in a very short time.

I cannot end this introduction without paying tribute to the memory of I. Bucur, who died of cancer in September 1976. Those who, like me, had the privilege of having him as a friend will not forget his exquisite kindness, his smiling humor, and the charm of his conversation.

Paris, 19 February 1977
Luc Illusie

Continuation anchor. The next SGA 5 translation unit begins with Exposé I, “Dualizing complexes”, in the same source file, after the volume bibliography and list of SGA volumes.

SGA 5, Exposé I
Dualizing Complexes

by A. Grothendieck
(written up by L. Illusie)

Draft English translation. This file starts the translation of SGA 5, Exposé I, following the volume introduction translated in Batch 001.

Introduction

This exposé is devoted to the biduality theorem in étale cohomology. The theory developed here and in SGA A XVIII (SGA A = SGA 4), for the étale topology and constructible torsion sheaves on preschemes, is formally very analogous to the theory developed in Hartshorne [H] for the Zariski topology and coherent sheaves; the latter served, to a large extent, as its model. But whereas, in the case of "continuous" coefficients, the existence of dualizing complexes presents no difficulty, in the case of "discrete" coefficients it is much more delicate to prove, and it seems to require resolution of singularities in an essential way. See, however, Deligne's biduality theorem (SGA $4^{1/2}$, Finiteness Theorem 4.3).

Paragraph 1 contains the definition of dualizing complexes and the formal properties that follow from it, in particular the interpretation of the dualizing functors as "interchangers of functors".

In paragraph 2, one shows that, on a connected locally noetherian prescheme, a dualizing complex is determined uniquely, up to translation of degrees and tensoring by an invertible sheaf.

In paragraph 3, one comes to the central theorem of the exposé (3.4.1), which asserts that, on a "good" regular prescheme X, for example an excellent noetherian regular prescheme of characteristic zero,[1] the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing.

Paragraph 4 gives applications of the theory to the local duality theorem. The latter says that, on a "good" prescheme X equipped with a closed point x whose residue field is separably closed, and for a constructible sheaf F, there is a perfect pairing between $\underline{\mathrm{H}}^i_x(F')$ and $\underline{\mathrm{H}}^{-i}_x(F)$, where F' is the dual of F, that is,

$$F' = \mathrm{R}\,\underline{\mathrm{Hom}}(F, K_X),$$

with K_X a dualizing complex.

Finally, paragraph 5 proves directly that, on a regular prescheme of dimension 1, the complex $(\mathbb{Z}/n\mathbb{Z})_X$ is dualizing, where n is prime to the residue characteristics.

1. Definition and formal properties of dualizing complexes

Fix a coefficient ring $A = \mathbb{Z}/n\mathbb{Z}$. If X is a prescheme, let A_X denote the constant sheaf on $X_{\text{ét}}$ with value A, and let $\mathrm{D}(X)$ denote the derived category of the category of A_X-modules. We also write $\mathrm{D}_c(X)$ for the full triangulated subcategory of $\mathrm{D}(X)$ consisting of complexes F such that $\mathrm{H}^i(F)$ is a constructible A_X-module for every i. Put

$$\mathrm{D}^*_c(X) = \mathrm{D}^*(X) \cap \mathrm{D}_c(X), \qquad * = \varnothing, +, -, \text{ or } b.$$

[1] This last restriction will become unnecessary once resolution of singularities à la Hironaka and the "purity theorem" are available for such preschemes.

Definition (1.1)**.** An A_X-module I is called *quasi-injective* if

$$\underline{\mathrm{Ext}}^i(F, I) = 0$$

for every constructible A_X-module F and every $i > 0$.

Remark (1.2)**.** a) Contrary to what happens in the case of "continuous" coefficients, cf. [H] II.7.17, a quasi-injective sheaf is not in general injective. For example, let $X = \mathrm{Spec}(\mathbb{R})$ and $A = \mathbb{Z}/2\mathbb{Z}$. The sheaf A_X is quasi-injective, but is not injective, nor even of finite injective dimension, since

$$\mathrm{H}^*(X; A_X) = \mathrm{H}^*(\mathbb{Z}/2\mathbb{Z}; \mathbb{Z}/2\mathbb{Z})$$

is a polynomial algebra on one generator of dimension 1.

b) Let I be a quasi-injective A_X-module. It is false in general that the relation $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > 0$, valid for constructible F, is valid for every F. Indeed take $A = \mathbb{F}_\ell$. Let k be a field, let $\overline{k}$ be a separable closure of k, let

$$f : \overline{x} = \mathrm{Spec}(\overline{k}) \longrightarrow X = \mathrm{Spec}(k)$$

be the canonical morphism, and suppose that, for every connected étale covering $X' \to X$, there exists a connected étale covering $X'' \to X'$ such that $[X'' : X']$ is divisible by ℓ; for instance, take $k = \mathbb{Q}$. Consider the exact sequence

$$(*) \qquad 0 \longrightarrow A_X \xrightarrow{\varepsilon} f_* f^* A_X \longrightarrow F \longrightarrow 0$$

defined by the adjunction arrow ε. In view of the hypothesis, the section of $\mathrm{Ext}^1(F, A_X)$ defined by $(*)$ vanishes over no étale U over X, and hence $\underline{\mathrm{Ext}}^1(F, A_X)_{\overline{x}} \neq 0$. On the other hand, one checks trivially that A_X is quasi-injective.

Proposition (1.3, cf. Hartshorne 1.7.6)**.** *Let X be a prescheme and let $N \in \mathbb{Z}$. The following conditions on $K \in \mathrm{ob}\, \mathrm{D}^+(X)$ are equivalent:*

(i) K is isomorphic, in $\mathrm{D}(X)$, to a bounded complex I such that each I^i is quasi-injective and $I^i = 0$ for $i > N$;

(ii) for every $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$ such that $\mathrm{H}^i(F) = 0$ for $i < 0$, one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$;

(iii) for every constructible A_X-module F, one has $\underline{\mathrm{Ext}}^i(F, K) = 0$ for $i > N$.

Proof. $(i) \Rightarrow (ii)$. Let $F \in \mathrm{ob}\, \mathrm{D}^b_c(X)$ be such that $\mathrm{H}^i(F) = 0$ for $i < 0$, and let $I \in \mathrm{ob}\, \mathrm{D}^+(X)$ be bounded, with each I^i quasi-injective and $I^i = 0$ for $i > N$. We show that $\underline{\mathrm{Ext}}^i(F, I) = 0$ for $i > N$. Arguing by induction on the number of indices i for which $\mathrm{H}^i(F) \neq 0$, one reduces, by dévissage, to the case where $N = 0$ and I is concentrated in degree 0. Then there is a biregular spectral sequence

$$E_2^{pq} = \underline{\mathrm{Ext}}^p(\mathrm{H}^{-q}(F), I) \Rightarrow \underline{\mathrm{Ext}}^*(F, I).$$

Since I is quasi-injective, $E_2^{pq} = 0$ for $p > 0$ and all q, and therefore

$$E_2^{0i} = \underline{\mathrm{Hom}}(\mathrm{H}^{-i}(F), I) \xrightarrow{\sim} \underline{\mathrm{Ext}}^i(F, I),$$

which gives the conclusion.

$(ii) \Rightarrow (iii)$ is trivial.

$(iii) \Rightarrow (i)$. We may suppose that K is bounded below and has injective components. Applying the hypothesis to $F = A_X$ gives $\mathrm{H}^i(K) = 0$ for $i > N$. Thus K may be replaced by an isomorphic complex K', bounded, with K'^i injective for $i < N$ and $K'^i = 0$ for $i > N$. One then sees that K'^N is quasi-injective, which completes the proof. □

Remark (1.4)**.** The redactor does not know whether (1.3) remains true when, in condition (ii), the hypothesis "$F \in \operatorname{ob} \mathrm{D}_c^b(X)$" is replaced by "$F \in \operatorname{ob} \mathrm{D}_c^+(X)$".

Definition (1.5)**.** The *quasi-injective dimension* of K, denoted $\dim . q. \operatorname{inj}(K)$, is the infimum, in $\overline{\mathbb{R}}$, of the integers N satisfying the equivalent conditions of (1.3).

If K has finite quasi-injective dimension, then the functor

$$\mathrm{R}\,\underline{\mathrm{Hom}}(\,\cdot\,, K) : \mathrm{D}(X) \longrightarrow \mathrm{D}(X)$$

induces a functor $\mathrm{D}_c^b(X) \to \mathrm{D}^b(X)$.

We shall see below that if X is smooth of dimension d over a field k and n is prime to the characteristic of k, then

$$\dim . q. \operatorname{inj}(A_X) = 2d.$$

The example of (1.2 a) shows that a complex of finite quasi-injective dimension need not be of finite injective dimension. However, one has the following result.

Proposition (1.6)**.** *Let X be a locally noetherian prescheme and let $K \in \operatorname{ob} \mathrm{D}^+(X)$. Suppose that there exists an integer N such that $\operatorname{cd}_{\mathbf{n}}(X) \leq N$, where $\mathbf{n}$ is the set of prime divisors of n (notation of SGA A X 1). Then, for $N' \in \mathbb{Z}$,*

$$\dim . q. \operatorname{inj}(K) \leq N' \quad \Longrightarrow \quad \dim . \operatorname{inj}(K) \leq N + N'.$$

The proof will use the following lemma.

Lemma (1.6.1)**.** *Let C be an abelian category satisfying (AB5) and possessing a family of generators $(U_i)_{i \in I}$. Let $K \in \operatorname{ob} \mathrm{D}^+(C)$ and $N \in \mathbb{Z}$. The following conditions are equivalent:*

(i) $\dim . \operatorname{inj}(K) \leq N$*;*

(ii) for every $i \in I$ and every quotient F of U_i, one has

$$\operatorname{Ext}^n(F, K) = 0 \qquad \textit{for } n > N.$$

Proof. It is enough to prove $(ii) \Rightarrow (i)$. By a well-known argument, cf. for example [H] 1.7.6, one is reduced to showing that if $K' \in \operatorname{ob}(C)$ is such that $\operatorname{Ext}^1(F, K') = 0$ whenever F is a quotient of some U_i, then K' is injective. In other words, one must prove that if every morphism from a subobject V of some U_i to K' extends to U_i, then K' is injective. This follows immediately from the proof of Lemma 1, p. 136 of Tohoku. □

Proof of (1.6). Suppose $\dim . q. \operatorname{inj}(K) \leq N'$ and prove that $\dim . \operatorname{inj}(K) \leq N + N'$. Since the category of A_X-modules has as generators the $A_{U,X}$, where U is étale of finite type over X, it is enough, by (1.6.1), to prove that

$$\operatorname{Ext}^i(X; F, K) = 0 \qquad \text{for } i > N + N'$$

when F is a quotient of such an $A_{U,X}$, hence constructible by SGA A IX 2.9. Consider the spectral sequence

$$E_2^{pq} = \mathrm{H}^p(X; \underline{\operatorname{Ext}}^q(F, K)) \Rightarrow \operatorname{Ext}^*(X; F, K).$$

The hypothesis $\operatorname{cd}_{\mathbf{n}}(X) \leq N$, respectively $\dim . q. \operatorname{inj}(K) \leq N'$, implies $E_2^{pq} = 0$ for $p > N$, respectively for $q > N'$. Thus $E_2^{pq} = 0$ for $p + q > N + N'$, whence the conclusion. □

Remark (1.7)**.** The hypothesis of (1.6) is satisfied, with $N = 2d$, if X is a prescheme of finite type and dimension d over a separably closed field k, with n prime to $\operatorname{char}(k)$; see SGA A X 4.

Proposition (1.8)**.** *Let $f : X \to Y$ be a quasi-finite separated morphism of noetherian preschemes, and let $K \in \operatorname{ob} \mathrm{D}^+(Y)$. Then, for $N \in \mathbb{Z}$,*

$$\dim . q. \operatorname{inj}(K) \leq N \quad \Longrightarrow \quad \dim . q. \operatorname{inj}(\mathrm{R}f^! K) \leq N.$$

Proof. By the Main Theorem (EGA IV 8.12.6), there exists a factorization $f = uf'$, where f' is an open immersion and u is finite. We have $\mathrm{R}f^! = \mathrm{R}f'^! \, \mathrm{R}u^!$, which reduces us to considering separately the case of an open immersion and the case of a finite morphism. If f is an open immersion, the assertion is trivial, since every constructible sheaf on X is then induced by a constructible sheaf on Y. Suppose then that f is finite, and let F be a constructible A_X-module. The duality isomorphisms (SGA A XVIII 3.1.9.6) give

$$f_* \underline{\operatorname{Ext}}^i(F, \mathrm{R}f^! K) \xrightarrow{\sim} \underline{\operatorname{Ext}}^i(f_* F, K).$$

Since $f_* F$ is constructible by SGA A IX 2.14, it follows that if $\dim . q. \operatorname{inj}(K) \leq N$, then

$$f_* \underline{\operatorname{Ext}}^i(F, \mathrm{R}f^! K) = 0 \qquad (i > N),$$

and hence $\underline{\operatorname{Ext}}^i(F, \mathrm{R}f^! K) = 0$ for $i > N$ by SGA A VIII 5.5. This proves the proposition. □

Let now X be a prescheme and let $K \in \operatorname{ob} \mathrm{D}^+(X)$. Denote by $\underline{D}_K$ the functor

$$\mathrm{R} \, \underline{\operatorname{Hom}}(\,\cdot\,, K).$$

We shall define, for $F \in \operatorname{ob} \mathrm{D}(X)$, a functorial homomorphism

$$F \longrightarrow \underline{D}_K \underline{D}_K F$$

(cf. [H] V.1.2). The construction works more generally on a ringed topos. One may first suppose that K is made up of injectives, which permits one to "remove the R". The identity morphism

$$\underline{\operatorname{Hom}}(F, K) \longrightarrow \underline{\operatorname{Hom}}(F, K)$$

defines, by Cartan's beloved formula, a homomorphism

$$F \otimes \underline{\operatorname{Hom}}(F, K) \longrightarrow K,$$

whence, by applying the same formula a second time, a homomorphism

$$F \longrightarrow \underline{\operatorname{Hom}}(\underline{\operatorname{Hom}}(F, K), K).$$

This is the announced homomorphism, and we call it the *canonical homomorphism.* We say that the pair (F, K) is *bidualizing*, or *satisfies biduality*, or also that F is *reflexive* relative to K, if the canonical homomorphism

$$F \longrightarrow \underline{D}_K \underline{D}_K F$$

is an isomorphism.

From now on, all preschemes considered will, unless expressly stated otherwise, be assumed locally noetherian.

Definition (1.9)**.** Let X be a prescheme. A complex $K \in \operatorname{ob} \mathrm{D}^+(X)$ is called *dualizing* if the following conditions are satisfied:

(i) K has finite quasi-injective dimension;

(ii) for every $F \in \operatorname{ob} \mathrm{D}_c^-(X)$, one has $\underline{D}_K(F) \in \operatorname{ob} \mathrm{D}_c^+(X)$;

(iii) every $F \in \operatorname{ob} \mathrm{D}_c^b(X)$ is reflexive relative to K.

Remark (1.10)**.** a) By a standard dévissage, truncating F and using the fact that $\mathrm{D}_c(X)$ is triangulated, one sees that condition (ii) is equivalent to:

(ii bis) for every constructible A_X-module F and every $i \in \mathbb{Z}$, the sheaf $\underline{\mathrm{Ext}}^i(F, K)$ is constructible.

b) Condition (ii bis) implies $K \in \operatorname{ob} \mathrm{D}_c^+(X)$. Contrary to the example of coherent sheaves (EGA 0_{III} 12.3.3), the converse may not be automatic. It does hold, however, see number 3, in the "good" cases, for example when one has resolution of singularities and purity on X, in particular if X is an excellent prescheme of characteristic zero.

c) By dévissage on F, one sees that condition (iii) is equivalent to:

(iii bis) every constructible A_X-module F is reflexive relative to K.

d) If $K \in \operatorname{ob} \mathrm{D}^+(X)$ satisfies (i) and (ii), respectively is dualizing, the functor $\underline{D}_K$ induces a functor, respectively an equivalence of categories,

$$(\mathrm{D}_c^b(X))^{\mathrm{op}} \longrightarrow \mathrm{D}_c^b(X).$$

e) If $K \in \operatorname{ob} \mathrm{D}^+(X)$ has finite *injective* dimension, cf. 1.6.1, and satisfies (ii), then for every $F \in \operatorname{ob} \mathrm{D}_c(X)$ one has $\underline{D}_K(F) \in \mathrm{D}_c(X)$, and $\underline{D}_K$ induces a functor

$$(\mathrm{D}_c^+(X))^{\mathrm{op}} \longrightarrow \mathrm{D}_c^-(X).$$

If, moreover, K is dualizing, every $F \in \operatorname{ob} \mathrm{D}_c(X)$ is reflexive relative to K, and $\underline{D}_K$ induces equivalences of categories

$$(\mathrm{D}_c(X))^{\mathrm{op}} \xrightarrow{\approx} \mathrm{D}_c(X), \qquad (\mathrm{D}_c^-(X))^{\mathrm{op}} \xrightarrow{\approx} \mathrm{D}_c^+(X), \qquad (\mathrm{D}_c^+(X))^{\mathrm{op}} \xrightarrow{\approx} \mathrm{D}_c^-(X).$$

It is not known whether this conclusion remains valid without the restriction that K have finite injective dimension.

We now turn to the
emphexchange formulae.

We shall not return to the notion of the Tor-dimension of a complex (SGA A XVII 4.1.9).[2] Let us retain only that, if X is a prescheme and $G \in \operatorname{ob} \mathrm{D}^-(X)$, the relation

$$\underline{\mathrm{Tor}}_i(F, G) = 0 \qquad (i > N)$$

for every A_X-module F is equivalent to the same relation with F constructible, thanks to SGA A IX 2.9 and to the fact that $\otimes$ commutes with direct limits. Thus there is no need to introduce a notion of "quasi-Tor-dimension".

Lemma (1.11, cf. Hartshorne II 4.3)**.** *a) Let X be a prescheme, let $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^-(X)$. Then*

$$F \otimes^{\mathrm{L}} G \in \operatorname{ob} \mathrm{D}_c(X)$$

if $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ or if G has finite Tor-dimension.[3]

[2]See also SGA 6 I 5.

[3]The first condition is repeated as in the working TeX source. This point should be checked against the scan during proofing.

b) Let $f : X \to Y$ be a morphism of preschemes, and let $G \in \operatorname{ob} \mathrm{D}_c^-(Y)$. Then $f^*G \in \operatorname{ob} \mathrm{D}_c(X)$.

Proof. Assertion b) follows trivially from SGA A IX 2.4. Let us prove a). Either of the hypotheses on (F, G) implies that the spectral sequence

$$E_2^{pq} = \sum_{q_1+q_2=q} \underline{\mathrm{Tor}}_{-p}(\mathrm{H}^{q_1}(F), \mathrm{H}^{q_2}(G)) \Rightarrow \mathrm{H}^*(F \otimes^{\mathrm{L}} G)$$

is biregular. We are therefore reduced to proving a) when F and G are concentrated in degree 0, that is, are constructible A_X-modules. Since the question is local, we may suppose X noetherian. Then F admits, by SGA A IX 2.7, a left resolution F', where the F'^i are of the form $A_{U,X}$ with U étale and of finite type over X. We have

$$F \otimes^{\mathrm{L}} G \xleftarrow{\sim} F' \otimes G,$$

and the conclusion follows because the $A_{U,X} \otimes G$ are constructible. □

Notation (1.11.1)**.** If X is a prescheme, we shall denote by $\mathrm{D}^b(X)_{\mathrm{torf}}$ (respectively $\mathrm{D}^b(X)_{\mathrm{qinjf}}$) the full subcategory of $\mathrm{D}(X)$ consisting of objects of finite Tor-dimension (respectively finite quasi-injective dimension).

Proposition (1.12, cf. Hartshorne V 2.6)**.** *Let X be a prescheme and let $K \in \operatorname{ob} \mathrm{D}^+(X)$.*

a) For $F \in \operatorname{ob} \mathrm{D}^-(X)$ and $G \in \operatorname{ob} \mathrm{D}^-(X)$, respectively for $F \in \operatorname{ob} \mathrm{D}(X)$ and $G \in \operatorname{ob} \mathrm{D}^b(X)_{\mathrm{torf}}$, there is a functorial isomorphism

$$\underline{D}_K(F \otimes^{\mathrm{L}} G) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, \underline{D}_K G).$$

b) If K satisfies conditions (i) and (ii) of (1.9), one has the implication

$$G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}} \quad \Longrightarrow \quad \underline{D}_K G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{qinjf}} \text{ and } \underline{D}_K G \text{ satisfies (1.9)(ii).}$$

c) If K is dualizing, there is, for $F \in \operatorname{ob} \mathrm{D}_c^-(X)$ and $G \in \operatorname{ob} \mathrm{D}_c^b(X)$, respectively for $F \in \operatorname{ob} \mathrm{D}(X)$, $G \in \operatorname{ob} \mathrm{D}_c^b(X)$, and $\underline{D}_K G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}}$, a functorial isomorphism

$$\underline{D}_K(F \otimes^{\mathrm{L}} \underline{D}_K G) \xrightarrow{\sim} \mathrm{R}\,\underline{\mathrm{Hom}}(F, G).$$

Moreover one has the implication

$$\underline{D}_K G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}} \quad \Longrightarrow \quad G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{qinjf}} \text{ and } G \text{ satisfies (1.9)(ii).}$$

d) If K is dualizing and of finite injective dimension, the implications in b) and c) are equivalences.

Proof. a) is none other than Cartan's adjunction formula (SGA 6 I 7.4).

b) Let $G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{torf}}$. As F ranges over constructible A_X-modules, the cohomology objects of $F \otimes^{\mathrm{L}} G$ are constructible by (1.11) and vanish outside two fixed bounds independent of F. Therefore the same is true for the cohomology objects of $\underline{D}_K(F \otimes^{\mathrm{L}} G)$, and the result follows from a).

c) The first assertion follows from formula a), applied to $\underline{D}_K G$ in place of G, and from the reflexivity of G. The second is deduced by an argument analogous to b).

d) Let us prove, for example, the converse implication in b). Suppose that $\underline{D}_K G \in \operatorname{ob} \mathrm{D}_c^b(X)_{\mathrm{qinjf}}$ and that $\underline{D}_K G$ satisfies (1.9)(ii). Then, as F ranges over constructible modules, the cohomology objects of $\underline{D}_K(F \otimes^{\mathrm{L}} G)$ are constructible and vanish outside two fixed bounds. Thus the same is true for the cohomology objects of

$$\underline{D}_K \underline{D}_K(F \otimes^{\mathrm{L}} G),$$

but

$$F \otimes^{\mathrm{L}} G \xrightarrow{\sim} \underline{D}_K \underline{D}_K(F \otimes^{\mathrm{L}} G)$$

since K is dualizing and of finite injective dimension, cf. (1.10 e). Hence G has finite Tor-dimension. The converse implication in c) is proved similarly. □

Remark (1.12.1)**.** a) The redactor does not know whether assertion d) of (1.12) is valid under the sole hypothesis that K is dualizing.

b) One expresses the preceding proposition figuratively by saying that the dualizing functor $\underline{D}_K$ exchanges the functors $\otimes^{\mathrm{L}}$ and $\mathrm{R}\,\underline{\mathrm{Hom}}$, and also the notions of finite Tor-dimension and finite quasi-injective dimension. This exchange is, moreover, fully satisfactory only when K has finite injective dimension.

Proposition (1.13)**.** *Let Y be a noetherian prescheme, let $f : X \to Y$ be a morphism of finite type, and let $K_Y \in \mathrm{ob}\,\mathrm{D}^+(Y)$. Suppose that n is prime to the residue characteristics of Y, and put*

$$K_X = \mathrm{R}f^! K_Y \qquad \textit{(SGA A XVIII 3.1)}, \qquad \underline{D}_X = \underline{D}_{K_X}, \quad \underline{D}_Y = \underline{D}_{K_Y}.$$

a) There is, for $F \in \mathrm{ob}\,\mathrm{D}^-(X)$, a functorial isomorphism

$$\textit{(i)} \qquad \mathrm{R}f_* \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y\,\mathrm{R}f_* F.$$

If, moreover, K_X and K_Y are dualizing,[4] *there is, for $F \in \mathrm{ob}\,\mathrm{D}^b_c(X)$, a functorial isomorphism*

$$\textit{(ii)} \qquad \mathrm{R}f_! \underline{D}_X(F) \xrightarrow{\sim} \underline{D}_Y\,\mathrm{R}f_* F.$$

b) There is, for $F \in \mathrm{D}^-_c(Y)$, a functorial isomorphism[5]

$$\textit{(i)} \qquad \mathrm{R}f^! \underline{D}_Y(F) \xrightarrow{\sim} \underline{D}_X f^* F.$$

If, moreover, K_X and K_Y are dualizing, there is, for $F \in \mathrm{D}^b_c(Y)$, a functorial isomorphism

$$\textit{(ii)} \qquad f^* \underline{D}_Y F \xrightarrow{\sim} \underline{D}_X \mathrm{R}f^! F.$$

Proof. a)(i) is none other than the duality isomorphism (SGA A XVIII 3.1.9.6).

a)(ii) is obtained by applying $\underline{D}_Y$ to the two sides of a)(i), written for the argument $\underline{D}_X F$, and by using the finiteness theorem (SGA A XVII 5.3.6), which asserts that

$$\mathrm{R}f_* \underline{D}_X(F) \in \mathrm{ob}\,\mathrm{D}_c(Y).$$

b)(i) is none other than the "induction formula" (SGA A XVIII 3.1.12.2).

b)(ii) is obtained by applying $\underline{D}_X$ to the two sides of b)(i), written for the argument $\underline{D}_Y F$, and using (1.12 b). □

Remark (Scholium 1.13.1)**.** One may say, figuratively, that the dualizing functors $\underline{D}_X$ and $\underline{D}_Y$ exchange the usual and unusual direct-image functors, respectively inverse-image functors:

$$\mathrm{R}f_* \quad \text{and} \quad \mathrm{R}f_!, \qquad f^* \quad \text{and} \quad \mathrm{R}f^!.$$

[4] We shall see below that, under fairly general conditions, K_Y dualizing implies K_X dualizing.

[5] If the functor $\mathrm{R}f^!$ has finite cohomological dimension, as is the case for example when Y is of finite type over a field (SGA A X 4.3 and XVIII 3.1.7), the isomorphism b)(i) is valid for $F \in \mathrm{ob}\,\mathrm{D}_c(Y)$.

Corollary (1.14)**.** *Under the hypotheses of 1.13, suppose further that f is proper, and let $F \in \operatorname{ob} \mathrm{D}^-(X)$. If (F, K_X) is bidualizing, then $(\mathrm{R}f_*F, K_Y)$ is bidualizing. Conversely, if f is finite and $(\mathrm{R}f_*F, K_Y)$ is bidualizing, then (F, K_X) is bidualizing.*

Proof. Suppose that (F, K_X) is bidualizing. Applying $\mathrm{R}f_*$ to the canonical isomorphism $F \xrightarrow{\sim} \underline{D}_X^2 F$, one obtains

$$\begin{aligned} \mathrm{R}f_*F &\xrightarrow{\sim} \mathrm{R}f_*\underline{D}_X\underline{D}_X F \\ &\xrightarrow{\sim} \underline{D}_Y \mathrm{R}f_* \underline{D}_X F \qquad \text{by a)(i)} \\ &\xrightarrow{\sim} \underline{D}_Y\underline{D}_Y \mathrm{R}f_* F \qquad \text{by a)(i).} \end{aligned}$$

The courageous reader will convince himself that the isomorphism obtained is the canonical one; this proves the first assertion.

Now suppose f finite and (f_*F, K_Y) bidualizing, since $\mathrm{R}f_* = f_*$ for f finite. We prove that the canonical morphism

$$F \longrightarrow \underline{D}_X^2 F$$

is an isomorphism. By Lemma 1.14.1 a) below, it is enough to show that the induced morphism

$$f_*F \longrightarrow f_*\underline{D}_X^2 F$$

is an isomorphism. But, by applying a)(i) twice, one convinces oneself that this latter morphism is none other than the canonical isomorphism

$$f_*F \longrightarrow \underline{D}_Y^2 f_*F,$$

which completes the proof. □

Lemma (1.14.1)**.** *Let $f : X \to Y$ be a finite morphism of preschemes.*

a) Let $u : F \to G$ be an arrow of $\mathrm{D}(X)$. Then u is an isomorphism if and only if f_*u is an isomorphism.

b) Let $F \in \operatorname{ob} \mathrm{D}(X)$. Then $F \in \operatorname{ob} \mathrm{D}_c(X)$ if and only if $f_*F \in \operatorname{ob} \mathrm{D}_c(Y)$.

Proof. a) Since the functor f_* is exact, one may restrict to the case where F and G are concentrated in degree zero. The assertion then follows immediately from the computation of the fibers of f_*F, respectively f_*G, in SGA A VIII 5.5.

b) One may again restrict to the case where F is concentrated in degree zero. Necessity is known by SGA A IX 2.14. Let us prove sufficiency, that is, that f_*F constructible implies F constructible. The question is local above, and hence a fortiori below, so we may suppose Y, and therefore X, affine. There then exists, by EGA IV 17.16.6, a finite family (Y_i) of affine subpreschemes of Y, pairwise disjoint and with union Y, such that if $X_i = X \times_Y Y_i$, the induced morphism

$$f_i : X_i \longrightarrow Y_i$$

factors as

$$X_i \xrightarrow{u_i} X_i' \xrightarrow{v_i} Y_i,$$

where u_i is finite, radicial, and surjective, and v_i is finite étale. Put $F|_{X_i} = F_i$. By the base-change theorem for a finite morphism (SGA A VIII 5.5), one has

$$f_{i*}F_i \xrightarrow{\sim} (f_*F)|_{Y_i},$$

and one is reduced, by SGA A IX 2.8, to proving the assertion for (f_i, F_i). Put $F_i' = u_{i*}F_i$, so that $f_{i*}F_i = v_{i*}F_i'$. It is clear from the definition of constructibility, respectively follows from SGA A VIII 1.1, that F_i, respectively F_i', is constructible if and only if F_i', respectively $v_{i*}F_i'$, is constructible. This gives the desired conclusion. □

Corollary (1.15)**.** *Let $f : X \to Y$ be a quasi-finite separated morphism of noetherian preschemes. Let $K_Y \in \operatorname{ob} \mathrm{D}^+(Y)$, and put, as in 1.13,*

$$K_X = \mathrm{R}f^! K_Y, \qquad \underline{D}_X = \underline{D}_{K_X}, \qquad \underline{D}_Y = \underline{D}_{K_Y}.$$

If K_Y is dualizing, then K_X is dualizing.

Proof. We already know from (1.8) that if K_Y has finite quasi-injective dimension, then so does K_X. Let us show that, if K_Y satisfies condition (ii), respectively (iii), of (1.9), then K_X satisfies it as well. By the Main Theorem (EGA IV 8.12.6), f factors as $f = f'i$, where i is an open immersion and f' is finite. Since

$$\mathrm{R}f^! \simeq \mathrm{R}i^! \, \mathrm{R}f'^!,$$

one must therefore consider separately the case of an open immersion and that of a finite morphism. If f is an open immersion, the assertion is trivial, since a constructible sheaf on X is induced by a constructible sheaf on Y, and the formation of $\mathrm{R}\,\underline{\mathrm{Hom}}$ commutes with restriction to an open. Suppose then that f is finite. We know from (1.14) that if K_Y satisfies (iii), then so does K_X, taking account of the finiteness theorem SGA A IX 2.14. It remains to prove that, if K_Y satisfies (ii), then so does K_X. Let $F \in \operatorname{ob} \mathrm{D}^-_c(X)$; we must show that $\underline{D}_X F \in \operatorname{ob} \mathrm{D}^+_c(X)$. By the finiteness theorem, $f_* F \in \operatorname{ob} \mathrm{D}^-_c(Y)$, hence by hypothesis $\underline{D}_Y f_* F \in \operatorname{ob} \mathrm{D}^+_c(Y)$. But

$$\underline{D}_Y f_* F \xrightarrow{\sim} f_* \underline{D}_X F$$

by (1.13 a)(i), and the conclusion follows from (1.14.1 b). $\square$

Continuation anchor. The next translated unit starts with Exposé I, Section 2, “Uniqueness of the dualizing complex”.